The Waste Collection Vehicle Routing Problem with Stochastic Time Dependent Demand Increments

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Copenhagen - Denmark, 01/07/2024

Summary

- Waste Collection and Variability
- Problem Formulation
- Preliminary Results
- Policies, Conclusions, and Future Works

Waste Management and Circular Economy

The European Green Deal establishes ambitious targets for reducing emissions within a medium-term timeframe.
European countries are committed to achieving carbon neutrality by 2050, with an interim goal of cutting emissions by 55% by 2030 compared to 1990 levels.
Circular economy practices and **optimized waste management** play a crucial role in achieving these sustainable development objectives[a].

[a] https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/story-von-der-leyen-commission/european-green-deal_en

Waste Collection and Stochastic Elements

- Overtourism





- Changing habits



- Growing (evolving) cities.

Waste Collection and Stochastic Elements



- Unespected demand (during shifts)
- Process (vehicles) disruption
- Extra shifts
- Need for new planning policies





The Problem addressed - vehicle routing problem with stochastic time-dependent demand increments

A homogeneous fleet of vehicles must visit a set of collection points with the aim of removing waste.

We divide the planning horizon (day) in not-overlapping time slots.

Each collection point must be visited exactly once during the planning horizon and it has a known initial demand which may increase along the day. The (uncertain) demand increments are revealed upon arrival at the collection points.

The goal is to minimize the expected total cost which also considers a penalty for the residual quantity of waste accumulated for the next day.

Stochastic Vehicle Routing

Team Orienteering - TOP

Waste collection routing problems with stochastic demand have been extensively studied in the literature, mostly considering demand unknown but stationary.

Markov et al.[a] deal with a demand varying on time. However, they consider a multi-period setting in which demands may increase from one period to another, but it is constant over the same period (routes span over a single period, then the exact quantity is know at the beginning of the daily plan).

Conversely, we study a single day setting, where the time-horizon is split in time-slots and routes can span over multiple time-slots.

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[a] I. Markov, M. Bierlaire, J.-F. Cordeau, Y. Maknoon, S. Varone, "Waste collection inventory routing with non-stationary stochastic demands", Computers & Operations Research 113, 104798 (2020)

Problem Formulation

Let m be the number of available vehicles and G a direct graph. Nodes $1, ..., n$ of G represent the collection points, whereas node 0 and nodes $n+1, ..., n+m$ represent the central depot (same location).

Sets:

- $V = \{0, 1, ..., n, n+1, ...n+m\}$: set of all nodes
- $V' = \{0, 1, ..., n\} = V \setminus \{n+1, ..., n+m\}$
- $V'' = \{1, ..., n, n+1, ...n+m\} = V \setminus \{0\}$
- $V''' = \{n+1, ...n+m\} = V \setminus V'$
- S : set of scenarios
- T : set of periods

The Problem - Parameters

C vehicle capacity

c_{ij} travel cost associated with arc (i, j), $i \in V'$, $j \in V''$ $(j \neq i)$

t_{ij} travel time associated with arc (i, j), $i \in V'$, $j \in V''$ $(j \neq i)$.

e^t starting time of period $t \in T$

ℓ^t ending time of period $t \in T$

p unit penalty cost for uncollected waste

d_j initial non-negative (deterministic) demand $j = 1, ..., n$

δ^s_{jt} non-negative demand increment $j = 1, ..., n$, (beginning) $t \in T$ $s \in S$

M, T_{max}, Q_{max} large constant

Formulation: Variables

x_{ij} : binary variable equal to 1 if a vehicle travels from node $i \in V'$ to node $j \in V''$ $(j \neq i)$, 0 otherwise (first stage routes)

U_{jt} : binary variable equal to 1 if node $j \in V''$ is visited in period $t \in T$

$\bar{T}_j$: arrival time at node $j \in V$

y_{ij}^s : binary variable equal to 1 if a vehicle skips arc (i, j) in scenario $s \in S$, 0 otherwise

Q_j^s : theoretical cumulative load at node $j \in V$ in scenario $s \in S$

R_j^s : quantity of waste exceeding the capacity at node $j = n + 1, ..., n + m$ in scenario $s \in S$

L_j^s : quantity of waste accumulated at collection point $j = 1, ..., n$ in scenario $s \in S$ in periods following its visit, and left over for the next day

ρ_j^s : binary variable equal to 1 if node $j = 1, ..., n$ is associated with event "capacity saturation" in scenario $s \in S$, 0 otherwise

α_j^s : binary variable equal to 1 if node j is the first node associated with event "capacity saturation" in scenario $s \in S$, 0 otherwise

Formulation: Objective Function

$$\min \quad \sum_{i \in V'} \sum_{j \in V''(j \neq i)} c_{ij} x_{ij} - \frac{1}{|S|} \sum_{s \in S} \sum_{i=1}^{n} \sum_{j \in V''(j \neq i)} c_{ij} y_{ij}^{s} + \frac{1}{|S|} \sum_{s \in S} \sum_{j=1}^{n} c_{j0} \alpha_{j}^{s} +$$

$$+ \frac{1}{|S|} \sum_{s \in S} \sum_{j=n+1}^{n+m} pR_{j}^{s} + \frac{1}{|S|} \sum_{s \in S} \sum_{j=1}^{n} pL_{j}^{s} \qquad (1)$$

Formulation: Constraints (1/4)

$$\sum_{j \in V'' \setminus \{i\}} x_{ij} = \sum_{j \in V' \setminus \{i\}} x_{ji} = \sum_{t \in T} U_{it}, \quad i = 1, ..., n \tag{2}$$

$$\sum_{j \in V''} x_{0j} = m \tag{3}$$

$$\sum_{j \in V'} x_{ji} = 1, \quad i = n+1, ..., n+m \tag{4}$$

$$\sum_{t \in T} U_{jt} = 1, \quad j \in V'' \tag{5}$$

$$\bar{T}_0 = 0 \tag{6}$$

$$\bar{T}_j \geq \bar{T}_i + t_{ij} - M(1 - x_{ij}), \quad i \in V', j \in V''(j \neq i) \tag{7}$$

Formulation: Constraints (2/4)

$$e^t - T_{max}(1 - U_{jt}) \leq \bar{T}_j \leq \ell^t + T_{max}(1 - U_{jt}), \quad j \in V'', t \in T \tag{8}$$

$$Q_0^s = 0, \quad s \in S \tag{9}$$

$$Q_j^s \geq Q_i^s - M(1 - x_{ij}) + d_j + \sum_{t \in T} \sum_{\tau=1}^{t} \delta_{j\tau}^s U_{jt}, \quad s \in S, i \in V', j = 1, ..., n \tag{10}$$

$$Q_j^s \leq Q_i^s + M(1 - x_{ij}) + d_j + \sum_{t \in T} \sum_{\tau=1}^{t} \delta_{j\tau}^s U_{jt}, \quad s \in S, i \in V', j = 1, ..., n \tag{11}$$

Formulation: Constraints (3/4)

$$Q_i^s - M(1 - x_{ij}) \leq Q_j^s \leq Q_i^s + M(1 - x_{ij}), \quad s \in S, i \in V''' \tag{12}$$

$$R_j^s \geq Q_j^s - C, \quad s \in S, j = n+1, ..., n+m \tag{13}$$

$$R_j^s \geq 0, \quad s \in S, j = n+1, ..., n+m \tag{14}$$

$$L_j^s = \sum_{t \in T} \sum_{\tau=t+1}^{|T|} \delta_{j\tau}^s U_{jt}, \quad s \in S, j = 1, ..., n \tag{15}$$

Formulation: Constraints (4/4)

$$\frac{Q_j^s - C}{Q_{max}} \leq \rho_j^s \leq 1 - \frac{C - Q_j^s}{Q_{max}}, \quad s \in S, j = 1, \ldots, n \tag{16}$$

$$y_{ij}^s \leq \frac{1}{2}(x_{ij} + \rho_i^s), \quad s \in S, i = 1, ..., n, j \in V''(j \neq i) \tag{17}$$

$$y_{0j}^s = 0, \quad s \in S, \quad j \in V'' \tag{18}$$

$$\alpha_j^s = \sum_{i \in V''} y_{ji}^s - \sum_{i \in V'} y_{ij}^s, \quad s \in S, \quad j = 1, \ldots, n \tag{19}$$

Example

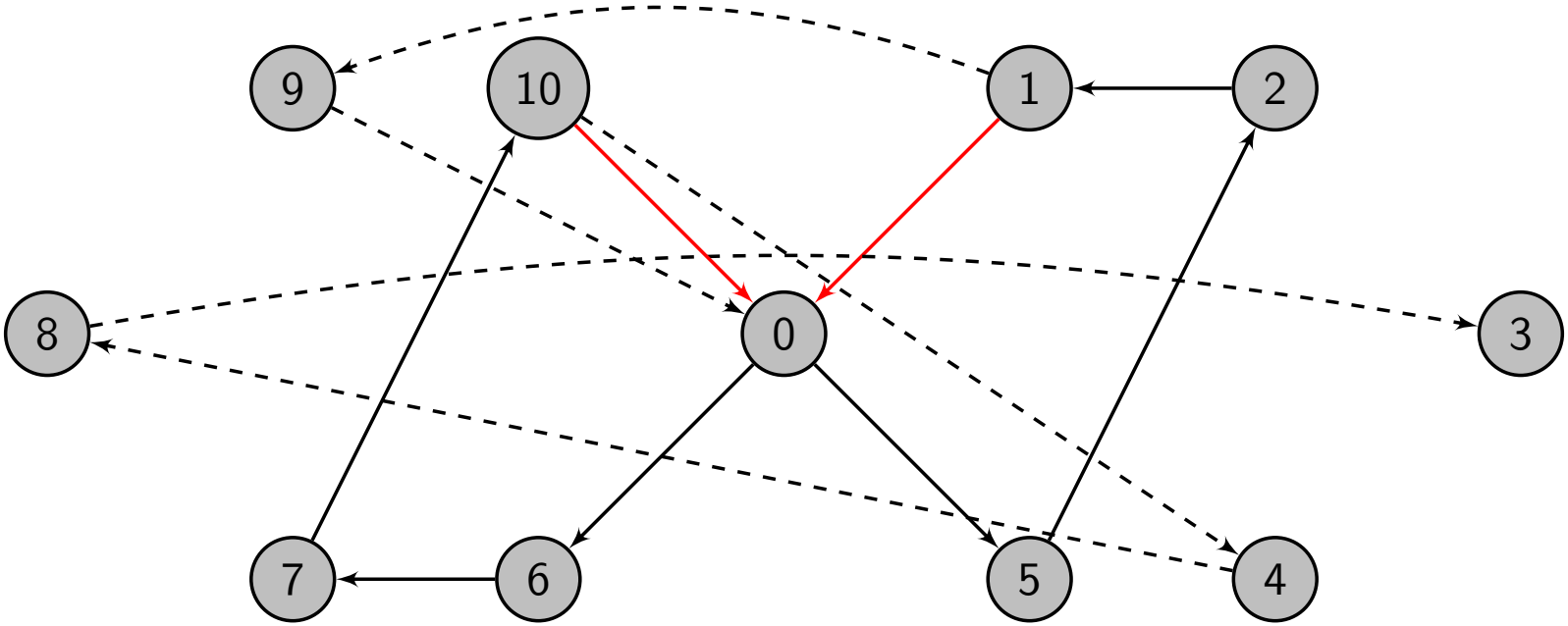


Figure: Dashed arcs consider $y^s_{i,j} = 1$, while red lines are the arcs connecting the vehicles to the depot

Computational Experiments: Some Instances

Table: Instances data

name	I1_S5	I1_S6	I1_S7	I1_S8
n	10	10	10	10
LT	3	3	3	3
m	3	3	3	3
LS	5	5	5	5
C	60	100	100	100
p	10	10	1	5
Tmax	100	100	100	100
Qmax	10000	10000	10000	10000
M	10000	10000	10000	10000
variability	medium	scattered	scattered	scattered

Computational Experiments: Some Results

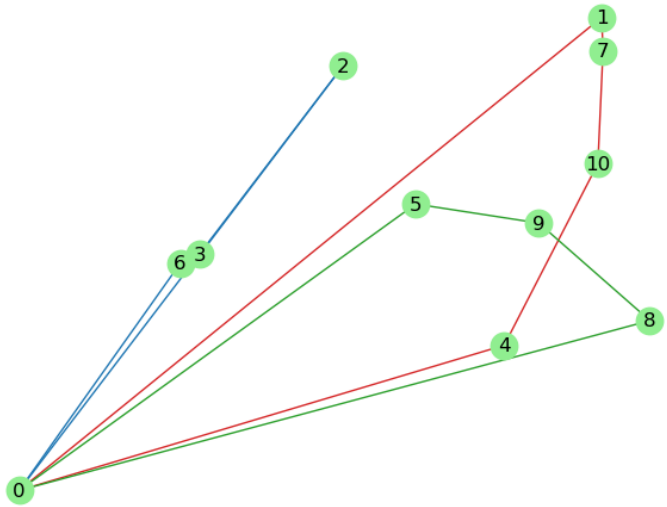
Table: Results

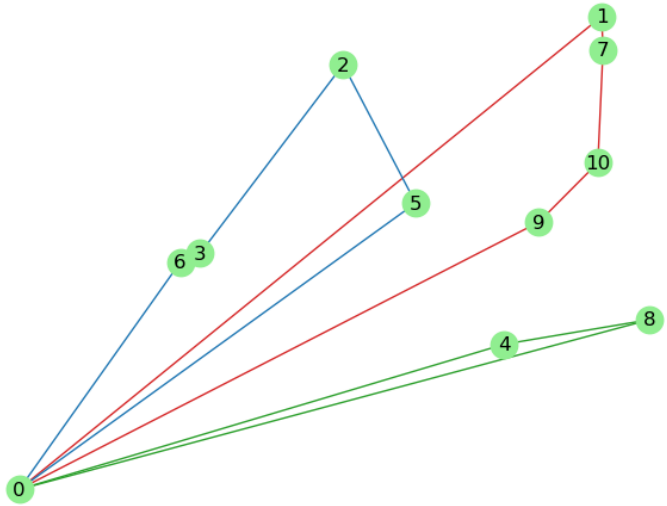
Instance	I1_S4	I1_S4_EEV	I1_S6	I_6_EEV	I1_S7	I1_S7_EEV	I1_S8	I1_S8_EEV
objValue	9552.03	9552.03	706.47	1004.74	131.42	173.70	387.00	603.35
runtime	10800	0.01	10307	0	10800	0	5652	0
gap	0.27	0.00	0.00	0.00	0.14	0.00	0.00	0.00
Z1	71.69	66.04	67.53	66.29	67.53	66.29	67.53	66.29
Z2	48.80	40.23	0.00	8.59	2.55	8.59	0.00	8.59
Z3	12.76	9.83	0.00	8.59	2.55	8.59	0.00	8.59
Z4	8803.04	8369.48	257.80	493.72	25.78	49.37	128.90	246.86
Z5	713.34	1146.90	381.14	444.74	38.11	58.04	190.57	290.21
Difference		0.00		42%		32%		56%

Solution comments

Instance S_6

- Stochastic
  - $0 \to 1 \to 7 \to 10 \to 4 \to 12$
  - $0 \to 2 \to 3 \to 6 \to 11$
  - $0 \to 8 \to 9 \to 5 \to 13$
- EEV
  - $0 \to 1 \to 7 \to 10 \to 9 \to 11$
  - $0 \to 5 \to 2 \to 3 \to 6 \to 13$
  - $0 \to 8 \to 4 \to 12$

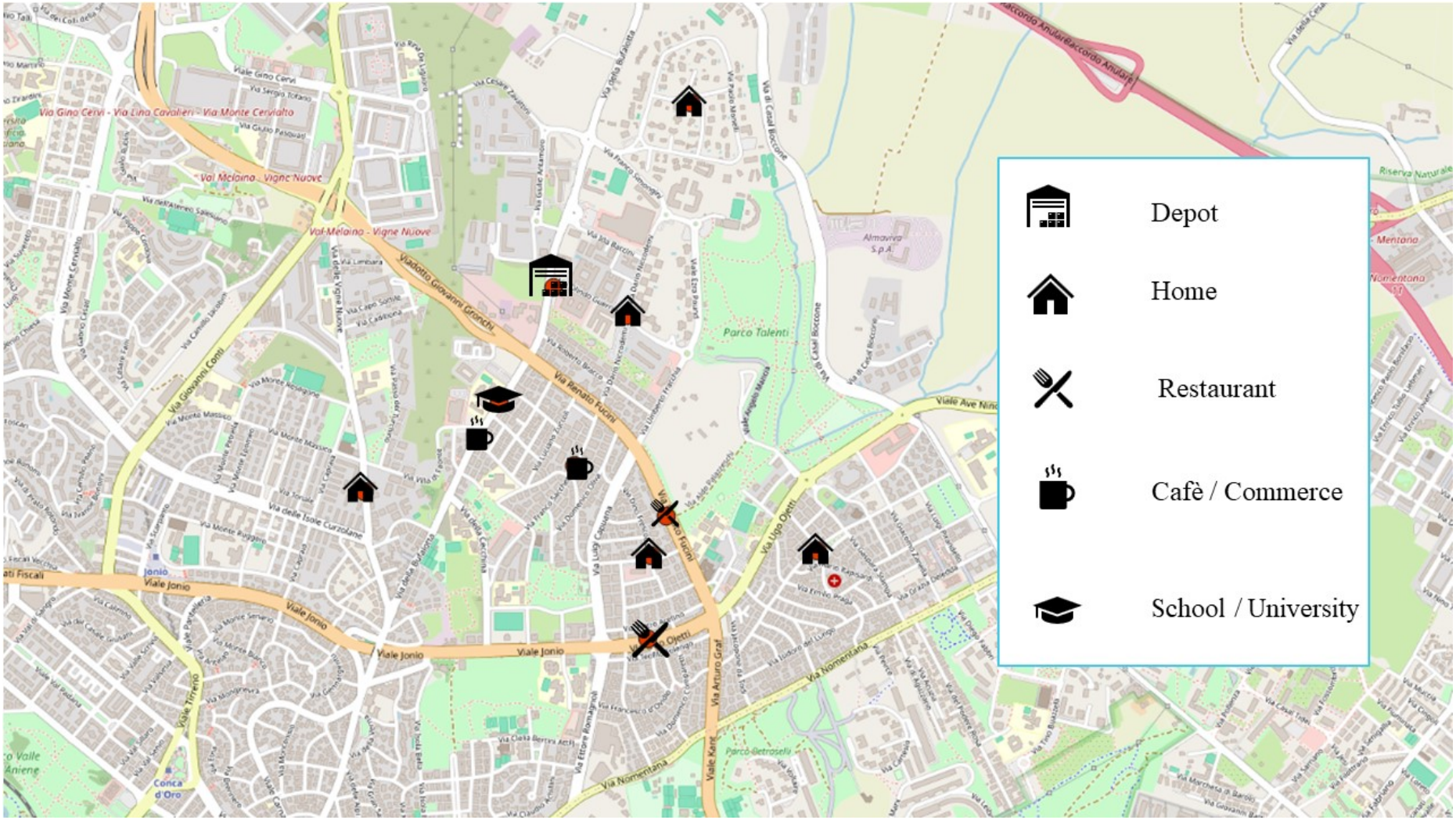




Realistic Instance - Different classes of customers

	demand	delta			
		8-12	12-16	16-20	20-24
home	medium	medium	low	low	medium
school	very low	low	medium	high	low
restaurant	high	very low	medium	very high	medium
bar	low	high	high	high	high

Realistic Instance



Realistic Instance: results and comments

Table: results realistic instance

Inst name	I1_R2	I1_R2_EEV
objValue	9496.31	10070.03
Z1	12607.92	11657.46
Z2	9677.65	5875.45
Z3	570.04	1444.02
Z4	3540.00	1794.00
Z5	2456.00	1050.00
Difference		6.04%

- different path / end of shift stratgy

Policies & Conclusion

- waste collection / waste management important for urban grow and sustainability
- cities evolves, dynamics and uncertainty to manage in order to offer proper services
- New stochastic model for waste collection
- Difficult to solve $\rightarrowtail$ policies can be designed.
  - lower capacity for vehicles
  - try to serve farthest users as last (fair?)
  - late collection

Conclusion

Thank You!

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